

Two Repertory Grids: NC State CSC and Its Rivals in AI

August 4, 2026 · Elements: NCSU CSC, 12 BOG-approved peers, Duke, and an ideal department scored from Provost Pfaendtner's published positions (Red Hat Research Quarterly, Summer 2026). Scale 1–5; 1 = ideal pole. Every Grid 1 cell is evidence-linked to public catalogs and program pages (see [evidence-grid1.md](#)).

1 Grid 1 — The Provost's Constructs

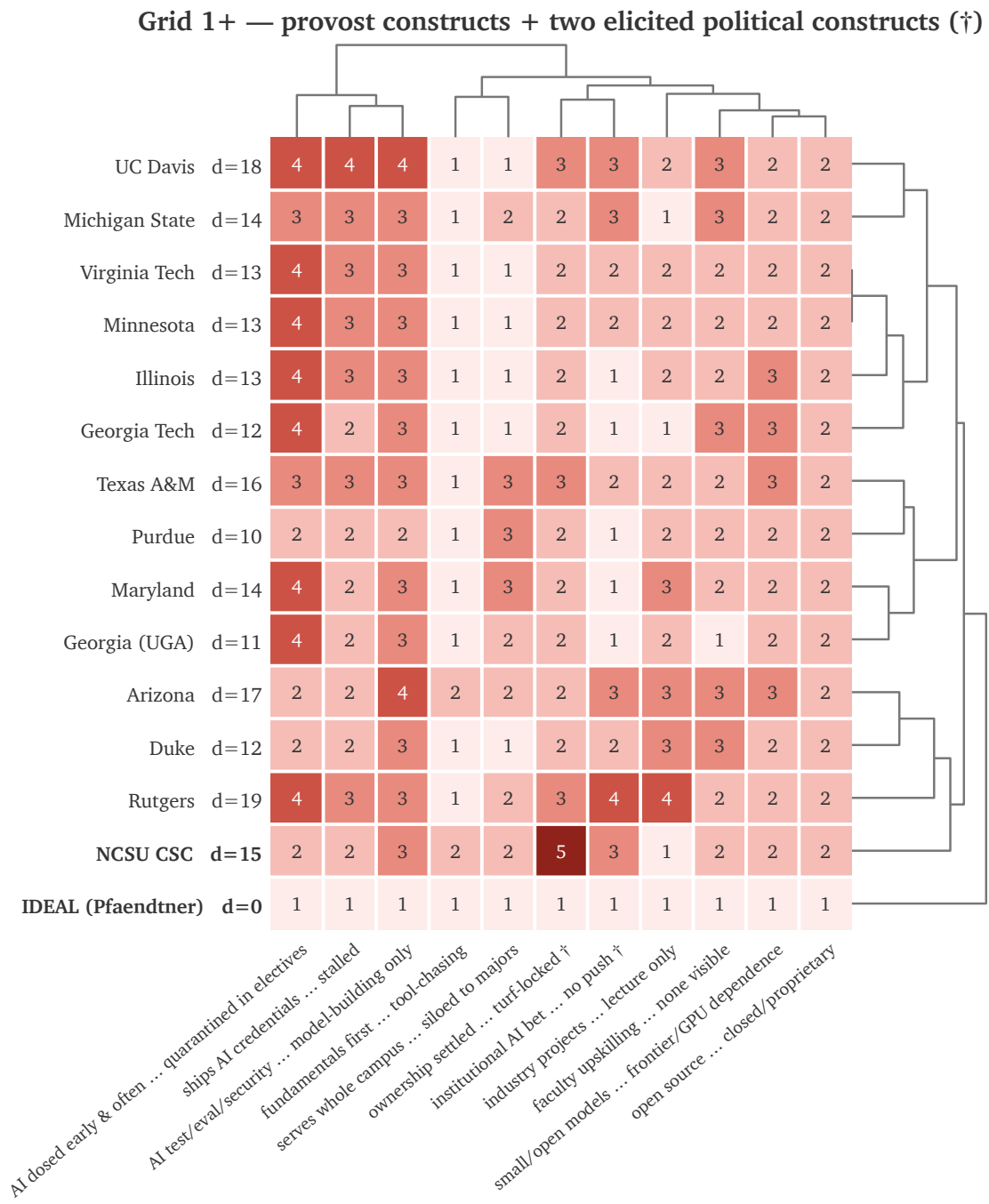
Nine bipolar constructs derived from Provost Pfaendtner's interview [1]; full pole names on the column labels below. Dendrograms: Ward linkage over Euclidean distance — constructs clustered above, departments clustered at right.

Findings

1. **By the Provost's own criteria, NCSU is already the closest department to his ideal** (tied with Purdue, $d = 9$). Drivers: applied-AI in year one (CSC 201), a thirty-year corporate-sponsored required capstone, live AI concentration and MCS track.
2. **Early AI dosing (C1) is the sector-wide failure:** nine of fourteen departments require no AI before senior electives. NCSU's 201/301 architecture answers the exact question the Provost posed in print.
3. **AI systems rigor (C9) is an empty lane:** no department requires testing, evaluation, or security-of-AI coursework; the content sits in graduate electives everywhere. The first mover defines the category.
4. Fundamentals (C2) offer no differentiation — the entire sector is already strong there.

2 Grid 1+ — Adding the Political Constructs

Triadic elicitation (six triads with T. Menzies) surfaced six candidate constructs; four were retired (brand duplicates public rankings, land-grant status is constant across the public peers, and two others carried no new variance). The two survivors — *AI ownership settled vs. turf-locked* and *institutional AI bet vs. no visible push* — are scoreable from the same public dossiers and are appended to Grid 1 (marked †).



Findings

1. **Two political constructs cost NCSU nine places:** first (of fourteen, $d = 9$) on the Provost's nine curricular constructs; tenth ($d = 15$) once turf-lock (5, worst in set) and institutional-bet (3) enter. The entire drop is politics — nothing in a classroom changed.
2. **The elicitation also failed one belief's evidence check:** UC Davis was construed alongside Purdue as “shipped AI degrees,” but Davis has shipped nothing (AI major/minor still at survey

stage). Purdue is the lone shipping benchmark.

3. **Where the two lenses agree:** shipping credentials and cross-campus reach matter under both. Safe load-bearing claims for any public document.
4. **Prescription follows the arithmetic:** the deficit is settled ownership and visible institutional commitment, not curriculum. Fix by exiting the ownership fight (service teaching claims no turf) and attaching to already-funded institutional signals (fellowship cohorts, applied-AI initiative).

3 Appendix — Construct Rubric

Each construct is bipolar; cells score 1 (left pole) to 5 (right pole). Anchors below fixed *before* scoring; every Grid 1 cell carries a one-line evidence note and URL in `evidence-grid1.md`. Constructs marked † were elicited by triadic comparison; the rest derive from Provost Pfaendtner’s RHRQ Summer 2026 interview (page references to the article PDF [1]).

	Score 1 anchor	Score 5 anchor	Source
C1	required AI touchpoint in year 1, reinforced in ≥ 2 later required courses	AI reachable only via senior/grad electives	Pfaendtner: “where does the dose of AI come early on...keep redosing it” (p. 10)
C2	theory/algorithms required before tool use	vendor-tool/prompting-led curriculum	Pfaendtner: “can’t replace the time...to learn fundamentals and critical thinking” (p. 10)
C3	courses run small/local/open models, cost-aware	teaching depends on paid frontier APIs / large clusters	Pfaendtner: “parsimony—small language models...transformational...the way we educate” (p. 17); Menzies EZR (arXiv 2606.03640)
C4	required projects with external/industry clients	lecture/exam dominant	article title “Stop talking, start doing”; p. 15
C5	open tools required; public course repos	proprietary stack, closed materials	Pfaendtner: “open source means we publish our work first” (p. 13–14)
C6	named faculty AI-upskilling program (fellowship/sabbatical)	none visible	Pfaendtner: faculty fellowship program (p. 11)
C7	CS-run AI service courses/minor open to all majors	AI courses restricted to CS majors	Pfaendtner: 1,850 freshmen—“How are we teaching them to think about AI?” (p. 9)
C8	AI BS + minor + MS live and enrolling	no AI-named credential	Pfaendtner: “being agile is essential” (p. 9)
C9	required testing/eval/security-of-AI coursework	model-building content only	advisory notes (Will, Dominic); Pfaendtner–Griffin exchange on debugging/security (p. 10)
C10†	one unit clearly owns AI credentials; campus accepts	proposals blocked/reconfigured by inter-unit contest	elicited, triad 5 ({UMD, Minn.} vs. NCSU); scored from dossier ownership facts
C11†	named university AI investment with money (institute, hiring wave, facility)	no visible institutional AI investment	elicited, triad 6 ({GT, Illinois} vs. Rutgers); scored from dossier partnership evidence

Score 3 = midpoint of the anchors (e.g. C1: AI required once; C10: parallel credentials, mild contest). No-public-signal cells score 3 and are flagged, never silently guessed. IDEAL scores 1 on every construct. Retired elicited constructs: brand (duplicates public rankings), land-grant status (constant across the public peers).

References

- [1] S. Griffin (interviewer), “Stop talking, start doing: how universities rise to the challenge of the AI revolution.” Interview with NC State Provost Jim Pfaendtner. *Red Hat Research Quarterly*, Vol. 8.1, Summer 2026, ISSN 2691-5278. <https://research.redhat.com/blog/article/stop-talking-start-doing-how-universities-rise-to-the-challenge-of-the-ai-revolution/>

- [2] UNC System Office, “2025 UNC Peer Institution Study,” approved by the Board of Governors May 15, 2025 (source of the 12 official NC State peers). <https://isa.ncsu.edu/facts-comparisons/peer-universities/>
- [3] T. Menzies and S. Srinivasan, “Can AI be Easy? Lessons Learned from the EZR.py Toolkit,” 2026. <https://arxiv.org/pdf/2606.03640>
- [4] G. A. Kelly, *The Psychology of Personal Constructs*, Norton, 1955 (repertory-grid method).

Method: repertory grid (Kelly, Personal Construct Theory). Grid 1 constructs supplied from Provost Pfaendtner’s RHRQ Summer 2026 interview; Augmented constructs (†) elicited from T. Menzies by triadic comparison; ratings scored by analyst from the same evidence dossiers. Distance = $\sum_i |x_i - \text{ideal}_i|$. Full rubric, anchors, and per-cell evidence URLs: `rubric-grid1.md`, `evidence-grid1.md`.